ComfyUI + Qwen Image 2.1: what everything actually does

Plain English. No definitions that define nothing. Every part and setting gets the same three answers.

What is it?
In words, not more acronyms.

What does it do?
To your picture, or your wait.

What do I set it to?
A real starting value.

Sized for `16 GB VRAM` `32 GB RAM` with notes for smaller cards

1 The big picture READ THIS FIRST

Qwen Image 2.1 does not paint the way you would. It starts with a screen of pure static and cleans it up a little at a time until it looks like your prompt. The same model does two jobs: make a picture from words, or change a picture you hand it.

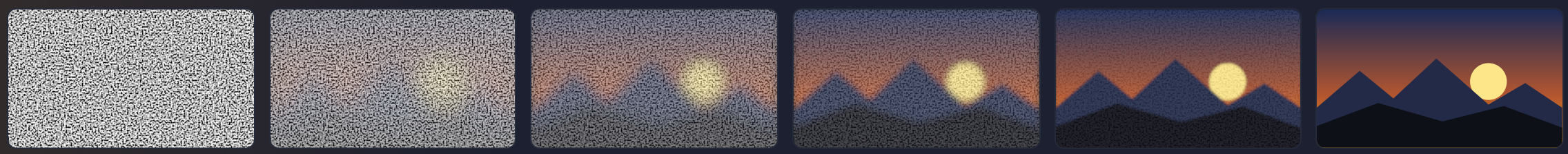
YOU
Prompt
Your words, plus one or more pictures if you are editing.

THE READER
Text encoder
Reads the prompt, looks at your pictures, and turns it all into numbers the artist can follow. Runs once per prompt.
`Load CLIP`

THE ARTIST
Diffusion model
Starts from static and cleans it up one step at a time, steered by the reader's numbers. Nearly all of your wait is here.
`Load Diffusion Model`

THE UNPACKER
VAE
The artist works on a small compressed version. The VAE turns it back into real pixels, transparency included.
`Load VAE`

RESULT
PNG
Up to 2048 x 2048, with a see-through background if you asked for one.



`start: static` `step 4` `step 8` `step 12` `step 16` `step 20: done`

A step is one cleanup pass. More passes, cleaner result, longer wait. (A drawing of the idea, not real output.)

Everything else on this page is one of three things: **a part** of that assembly line, **a file** that one of the parts loads, or **a knob** on the artist.

2 What it does ONE MODEL, FIVE JOBS

There is no mode switch. **What you plug in and what you ask for decides the job.**

JOB	YOU GIVE IT	WHAT IT DOES WITH IT	REACH FOR IT WHEN
Text to image	A prompt. Nothing else.	Invents the whole picture. It is unusually good at readable text inside the image: signs, labels, posters, page layouts.	You are starting from nothing, or you need words in the picture to actually be spelled right.
Image edit	A prompt and one picture, as `image_1`.	Changes what you ask for and leaves the rest alone. The output keeps the size and shape of your picture.	You like a picture and want one thing different: the outfit, the background, the lighting, an object removed.
Edit with references	A prompt and up to 10 pictures.	`image_1` is the picture being edited. The others are things to borrow from: a shirt, a face, a product, a style.	You want to combine things. "Put the shirt from `<image2>` on the person in `<image1>`."
Transparent image	A prompt wrapped in one set sentence (section 7).	Makes the subject on a truly see-through background, not a white one. This is what RGBA means: the usual red, green and blue, plus an alpha channel for transparency.	You need a cut-out you can drop onto something else: a logo, a sticker, a product shot.
Background removal	One picture and the prompt "Remove the background, and output a PNG image".	Returns your subject with the background made transparent.	You already have the picture and only want the cut-out.

One file does all of it

Generating and editing use the same diffusion model. If nothing is plugged into the image inputs, it generates. If a picture is plugged in, it edits. ComfyUI ships a separate template for each job so you do not have to rewire anything.

Name your pictures in the prompt

The model only knows which picture you mean if you say so. Call them by tag, in the order they are plugged in: `<image1>` `<image2>` up to `<image10>`. The first one is always the picture being edited.

Say what stays

An edit prompt works best with two halves: what to change, and what to keep. "Keep the character and pose in `<image1>` unchanged" does as much work as the change itself.

3 The parts WHAT EACH FILE YOU LOAD IS FOR

A workflow loads **three model files**. Each one has its own loader box on the canvas.

Diffusion model THE ARTIST

`also called: checkpoint, UNet, DiT | field: unet_name`

WHAT IT DOES

Makes the picture. This **is** the model. About 7 billion numbers, which is small by current standards.

WHY YOU CARE

It works on every single step, so it sets your speed. Because it is small, it is one of the easier current models to run well on a mid-range card.

Text encoder THE READER

`also called: CLIP | field: clip_name`

WHAT IT DOES

Reads your prompt and looks at any pictures you plugged in, then hands the artist numbers that mean the same thing.

WHY YOU CARE

It is a full language and vision model (Qwen3-VL, 8 billion parameters) and it is **bigger than the artist**. It runs once per prompt, then steps aside. Change only the seed and ComfyUI reuses its last answer.

Set the loader's type to `qwen_image`.

VAE THE UNPACKER

`field: vae_name`

WHAT IT DOES

Turns the artist's compressed result into real pixels, and squeezes your input pictures down on the way in.

WHY YOU CARE

The 2.1 VAE carries four channels, not three. The fourth is transparency. Use `qwen_image_2.1_vae_bf16`. The older `qwen_image_vae` belongs to earlier Qwen Image models.

Optional parts you will see in the templates

Prompt enhancer THE REWRITER

A second language model that rewrites your short prompt into a long, detailed one before the reader sees it. It comes in two files: `pe_t2i` for making pictures and `pe_i2i` for edits. It costs another 6 to 9 GB to load and its own wait. Leave it off while you are learning, so the picture reflects what **you** typed.

Qwen Image 2.1 Cache A SPEED-UP FOR EDITS

During an edit, your prompt and reference pictures are the same on every step. This node lets the model work that part out once and reuse it. `device` says where the saved work is kept: `auto` uses spare VRAM, then RAM. `dtype` shrinks it: `int8` halves it at about the same accuracy. Leave both on their defaults unless you run out of memory.

Add-ons and words you will keep seeing

LoRA

A small patch file that sits on top of the artist and bends it toward something: a face, a style, a product, or speed. **Strength** is how hard it bends. 0 is off, 1.0 is what its maker intended, and well past that the picture overcooks. It has to be made for Qwen Image 2.1. LoRAs for older Qwen Image models are not a safe bet.

Turbo

A version taught to finish in 8 steps. It comes two ways: as its own diffusion model (`qwen_image_2.1_turbo`) or as a LoRA you add to the normal one. It is a recent addition, so update ComfyUI before looking for it.

Checkpoint

Just means a saved model file. People use it loosely for the diffusion model.

.safetensors

The container, like .zip or .mp4. It tells you nothing about what is inside. The name is literal: the file can only hold numbers, so it cannot hide a program the way the older .ckpt and .pt files could.

4 File names WHICH ONE TO DOWNLOAD

Read a model's filename left to right and it tells you exactly what you are getting.

`qwen_image` `_2.1` `_int8_convrot` `.safetensors`

qwen_image
The model family.

2.1
The version. Files for 2.1 only work with other 2.1 files. If you see `_turbo` after it, that is the 8-step version.

int8_convrot
How its numbers are stored. This is the part that sets the file size.

.safetensors
The container. Ignore it.

The one idea behind all the letters

A model is a giant list of numbers. The letters say **how many bits each number is stored with**. Fewer bits: smaller file, more of it fits on your card, slightly less exact result. Same trade as JPEG quality or MP3 bitrate.

bf16
16 bits per number. The original, full-quality file, and the biggest one.

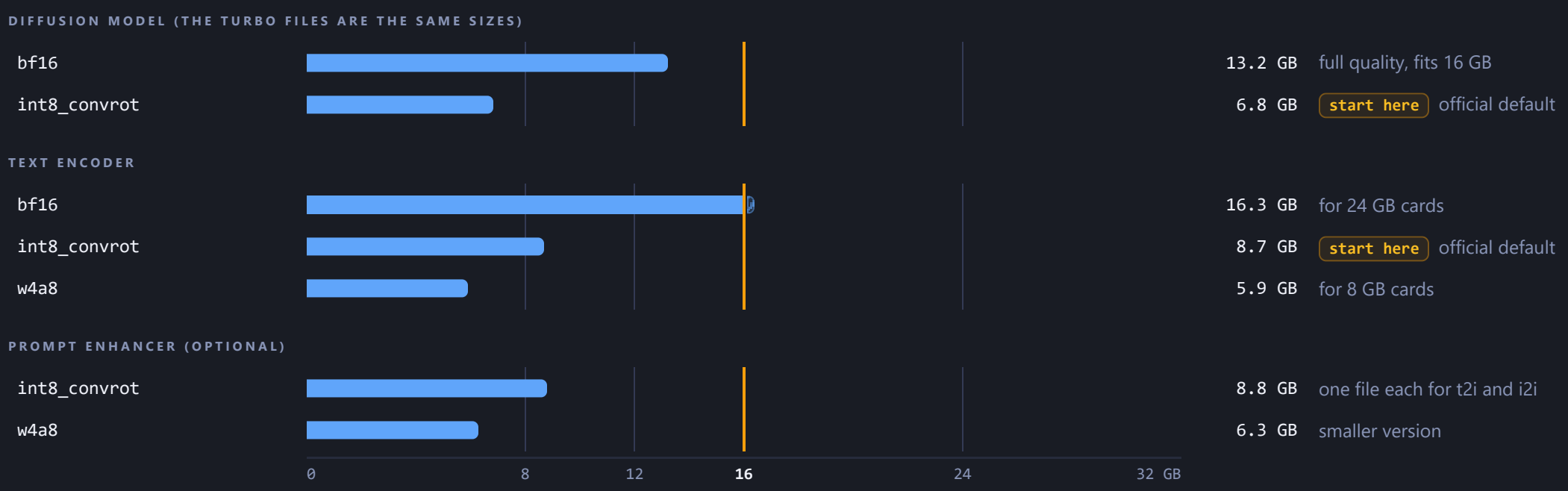
int8
8 bits per number, stored as whole numbers. About half the size of the 16-bit original.

convrot
The trick that makes int8 hold up. Before rounding, the numbers get shuffled so a few extreme values cannot wreck the precision of the rest. You set nothing. Read it as "the good int8". It wants an up-to-date ComfyUI.

w4a8
4-bit weights, 8-bit activations. Weights are the model's stored numbers. Activations are the numbers flowing through it while it works. Smaller again than int8, and a little rougher.

File size against your card

fits on a 16 GB card · past 16 GB, spills into system RAM · 16 GB · faint lines: 8, 12 and 24 GB



Sizes are the official files. A model also needs working room on the card while it runs, more of it at 2K than at 1K, so a file that sits right under your line is tighter than it looks.

Download this set

PART	FILE	SIZE	GOES IN COMFYUI/MODELS/
Diffusion model	qwen_image_2.1_int8_convrot.safetensors	6.8 GB	`diffusion_models`
Text encoder	qwen3vl_8b_int8_convrot.safetensors	8.7 GB	`text_encoders`
VAE	qwen_image_2.1_vae_bf16.safetensors	0.6 GB	`vae`

This is the exact set ComfyUI's built-in Qwen Image 2.1 templates ask for, and the same three files cover generating and editing. All of it is at huggingface.co/Comfy-Org/Qwen-Image-2.1. The model card lists the Qwen Research License, so read it before any commercial use.

Smaller is not automatically better. A smaller file helps for one reason: more of it fits on the card. Every step down in bits costs a little exactness. On a 16 GB card the default set already fits, so the upgrade worth trying is the `bf16` diffusion model. Compare it against the default with the same seed before you keep it.

5 Settings THE KNOBS ON THE ARTIST

Change **one at a time, with the seed fixed**, or you will never know which one did it.

KNOB	WHAT IT ACTUALLY DOES	TURN IT UP AND	START AT
Resolution megapixels, or width x height	How many pixels are in the picture. The template sets it as megapixels plus a shape (square, wide, tall), and works out width and height for you in multiples of 32.	The picture gets sharper and holds more small detail. Time and memory climb at least as fast as the pixel count: 2K has four times the pixels of 1K. The model's native ceiling is 2048 x 2048.	`1 MP = 1024 x 1024` Go to 4 MP (2048 x 2048) once it runs and you want the detail.

Seed	The exact pattern of static the artist starts from. It is a number that picks a pattern, nothing more.	There is no up or down, only different. Same seed with everything else the same gives you the same picture again. A new seed is a new take on the same prompt.	`fixed`
			While you test a setting.
			`random`
			When you just want more takes.

Steps	How many cleanup passes the artist gets.	Cleaner detail and better text. Time rises in a straight line: 50 steps takes twice as long as 25. Qwen's own reference setup uses 40 to 50. ComfyUI's template starts at 25 because that is usually enough.	`25` Go to 30 or more if small detail or lettering stays muddy.
CFG	How hard to push toward the prompt and away from the negative prompt. At 1 it is off. Above 1, the artist runs twice on every step, once for each.	The picture follows the prompt harder and every step takes twice as long. Past a point it overcooks: harsh contrast, burnt colour.	`1` This is the official path. Raise it, to about 2, only if you are using a negative prompt.
Negative prompt	A list of things you do not want in the picture.	Does not apply at the default. With CFG at 1 this box is ignored completely, however much you type in it.	`leave it empty` It only works once CFG is above 1.
Sampler	The method the artist uses on each pass to decide what to clean up next.	Not a more or less knob. A different sampler reaches a slightly different result in the same number of steps. Fancier ones can get there in fewer.	`euler` The official pick. Do not tune this early.
Scheduler	The pacing. It decides how much static comes off at each step: big chunks early, when the layout gets decided, and fine touches late.	Not a more or less knob either. Different pacing, same number of steps.	`simple` Leave it.
Turbo	The 8-step version of the model, loaded as its own file or as a LoRA.	Runs finish in about a third of the time. Expect to give up some fine detail for it.	`8 steps` Use it for trying prompts. Do the keeper on the normal model.
resolution edit workflows only	Not the output size. It is how big your reference pictures are when the model looks at them: about this many pixels on each side, keeping their shape.	The model sees more detail in your references, and each step costs more. 0 means look at each picture at its own size.	`1024` The official default. The model supports up to 2048.
custom_size edit workflows only	Off: the result is the same size as `image_1`. On: the result uses the size you set yourself.	Does not apply. If you turn it on, keep the size close to `image_1`, or the edit drifts.	`off` Let the picture you are editing set the size.
LoRA strength	How hard a LoRA bends the artist.	Stronger effect, then overcooked: waxy skin, harsh contrast, warped shapes.	`1.0` Then nudge by 0.1 or 0.2.

6 Your hardware WHY IT IS SLOW, AND WHAT THE ERRORS MEAN

This is one of the friendlier models to run. Every part of the default set is smaller than a 16 GB card.

VRAM THE WORKBENCH

The graphics card's own memory. Whatever is sitting on it gets worked on at full speed, so this is the number that decides what is comfortable. This guide assumes 16 GB.

RAM THE SHELF

The PC's regular memory. Model files wait here when it is not their turn, and anything that does not fit on the workbench lives here. This guide assumes 32 GB.

Offloading THE SHUFFLE

When a model is bigger than the workbench, ComfyUI keeps the overflow on the shelf and swaps pieces on and off the card as each is needed. It is automatic. The result comes out the same. It only costs time.

The numbers

The default set is about **16 GB of model** in total: artist 6.8, reader 8.7, unpacker 0.6. They do not all sit on the card at once. They **take turns**: the reader goes first and steps aside, the artist does its steps, the unpacker finishes. On 16 GB nothing has to spill. On 12 GB it still fits. On 8 GB the artist fits and the reader does not quite, which is what the `w4a8` reader is for.

Too slow, or out of memory? Pull these in order

1. **Lower the megapixels.** Back to 1. The biggest saving for one click.
2. **Fewer steps.** 25, or Turbo at 8. This saves time, not memory.
3. **Turn the prompt enhancer off.** It is a whole extra model to load and run.
4. **A smaller reader.** `qwen3vl_8b_w4a8` (5.9 GB) in place of the int8 one.
5. **For edits, shrink the cache.** Set `dtype` to `int8` on the Qwen Image 2.1 Cache node, and use fewer or smaller reference pictures.
6. **Free the machine.** Close everything else that uses RAM or the graphics card.

What an error is telling you

`Red boxes when a workflow opens`
Missing custom nodes, or the workflow needs a newer ComfyUI. Update first, then let ComfyUI Manager install what is missing.

`"Value not in list" on a loader`
That file is not in the right models folder, or it has a different name than the workflow expects. Pick yours from the dropdown.

`"CUDA out of memory" / "Allocation on device"`
The card ran out of VRAM. Work down the list on the left.

`Dies after the last step finished`
Usually the VAE running out while unpacking. Lower the resolution.

`The whole PC crawls or freezes`
RAM is full and Windows is using the disk as memory. Close other programs, or move to smaller files.

The other words in this corner

CUDA / PyTorch THE PLUMBING

PyTorch is the math engine ComfyUI is built on. CUDA is NVIDIA's layer that lets PyTorch use the graphics card. Both come bundled with ComfyUI and you never touch them. You only meet them in messages: "CUDA out of memory" means the card is full, and a tag like `cu130` in the startup log is just the CUDA version PyTorch was built for.

Custom node AN ADD-ON

Every box on the canvas is a node. Custom nodes are boxes written by other people, installed through ComfyUI Manager. **Qwen Image 2.1 itself needs none**: its nodes are built into ComfyUI. You only need a pack when a workflow you downloaded uses one. Each pack is code running on your PC, so stick to well-known ones.

rgthree A POPULAR PACK

A set of convenience nodes that shows up in a lot of shared workflows. The two you will meet most: a LoRA loader that stacks several LoRAs in one box, and switches that turn whole groups of a workflow on and off.

7 First run A KNOWN-GOOD STARTING POINT

Get one plain picture working with the official setup before you touch anything clever.

The recipe

1. **Update ComfyUI.** Support for 2.1 is recent, and older builds cannot open the templates.
2. **Open the built-in template.** Workflow menu, Browse Templates, search for Qwen Image 2.1, pick text to image.
3. **Get the files.** ComfyUI lists what is missing and where it goes. It is the set from section 4.
4. **Leave the defaults.** 1 megapixel, `25` steps, CFG `1`, euler, simple.
5. **Run it.** The first run is the slowest, because every file has to come off the disk.
6. **Then try an edit.** Open the image edit template. It uses the same three files.

The prompt

Making a picture. Long and specific beats short and vague: subject, setting, lighting, camera, style. For words inside the picture, put the exact text in quotes.

Editing. Say what changes and what stays, and name the pictures: "Keep the character and pose in `<image1>` unchanged, put the shirt from `<image2>` on the character."

Transparent background. Wrap your description in this exact sentence, then save as PNG:

```
This is an RGBA format image with transparency. [your description]. The image has an alpha channel and a transparent background.
```

Checked against: ComfyUI Qwen Image 2.1 docs, Comfy-Org/Qwen-Image-2.1 file list, Qwen Image 2.1 model card, ComfyUI's built-in nodes and templates. October 2026.